

Cybernetic Security

Wireless Communication Security

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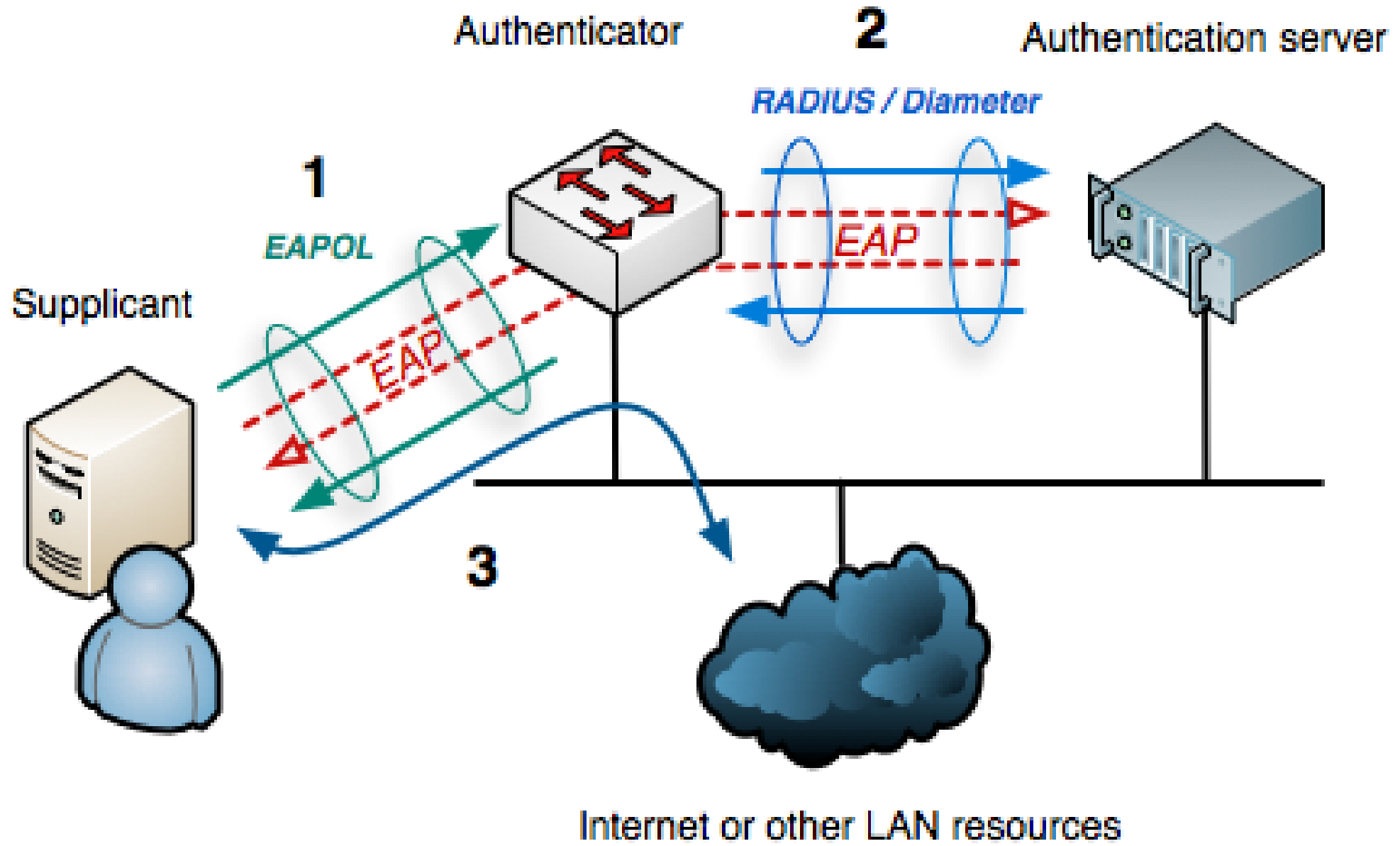
Lecture Agenda

IEEE 802.1x
WPA: PSK vs Enterprise
Wireless controllers
Radio spectrum monitoring
IoT networks

IEEE 802.1x

- Standard for authentication&authorization devices when accessing a network:
 - Wired networks: authenticating switch ports,
 - Wireless: authentication on the access point.
 - Defines encapsulation of EAP into L2 frames (EAPOL – EAP over LAN).
 - Authentication server (e.g. RADIUS) is required.

IEEE 802.1x Diagram



Wi-Fi Protected Access (WPA)

- WPA (2003):
 - TKIP encryption – temporary solution for prompt replacing of WEP on endpoints.
- WPA2 (2004):
 - implementing IEEE 802.11i.
 - AES support added.
- WPA3 (2018):
 - PSK is replaced with Key-exchange mechanism Simultaneous Authentication of Equals – according to IEEE 802.11s.

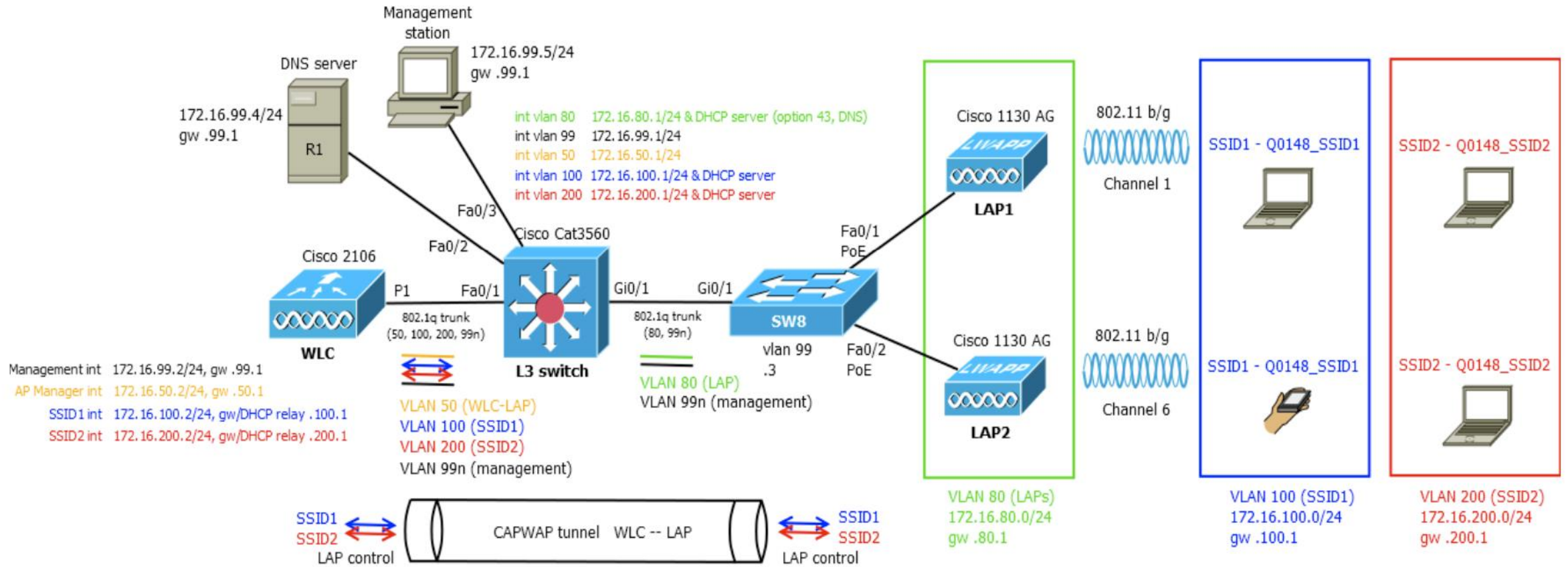
WPA Modes

- For small WiFi network:
 - Pre-Shared Key (PSK),
 - In WPA3 – replaced with SAE.
 - No authentication server is needed in LAN.
 - Shared passphrase (8 chars min.) transforms into 256-bit PSK (SSID as salt).
 - Encryption key for a device: 128b derived from PSK.
- Larger WiFi network (in a corporation/organization):
 - Central authentication server is available (e.g. RADIUS):
 - Combination with IEEE 802.1x is used (“enterprise WPA” or “WPA-802.1X mode”).
 - EAP protocol is used for authentication.

Wireless AP Central Management

- Small WiFi network:
 - 2-10 AP,
 - managed separately, including:
 - Each SSID, security settings, MAC address filters, admin passwords, etc.
 - For frequent changes, it is not flexible enough.
- Larger WiFi network:
 - Central management is necessary:
 - WireLess Controller (WLC):
 - software process (running on one of APs), or
 - hardware appliance (like a server in LAN).
 - Allows central configuration of:
 - SSID incl. security setting,
 - Optional MAC filtering etc.
 - Radio signal parameter measurement:
 - S/N, attenuation, interference, crosstalk for every channel,
 - Rogue AP detection.
 - Dynamic channel change (coordinated across all APs in the network).

Wireless Controller – Topology Example



Radio Spectrum for WiFi Networks

- ISM (2.4 GHz) frequency range:
 - Longest range,
 - Worse transmission conditions (interference channel overlap).
- 5 GHz range
 - Came later,
 - No channel overlap
 - More channels.
- 6 GHz and other ranges:
 - Newly added,
 - Still not completed concessions in some countries.

Radio Spectrum in WiFi: Potential Issues

- High attenuation:
 - too many obstacles (walls, ...),
 - too long distance,
- Bad S/N,
- Interference with other sources of radio signal:
 - incl. microwave ovens, remote controls, cell phones, ...
- Unauthorized (rogue) AP:
 - Employee brought AP without permission,
 - Attacker installed an AP nearby.
 - Moreover, the SSID could be identical (e.g. eduroam).

Rogue AP Detection

- WiFi AP not under control of the (corporate) network administrator:
 - Could compromise security measures for accessing the network (e.g. DHCP),
 - Could eavesdrop the private network traffic:
 - By luring clients to connect to it (instead of legitimate APs).
 - Could flood the network with traffic (DoS),
 - Could interfere with radio signal by legitimate Aps causing signal attenuation,
 - Could mess up a channel resulting in forced channel number change,
 - ...
- Advanced WLC can detect them:
 - Locate them precisely,
 - Sometimes even eliminate their impact before they are removed.

Internet of Things Networks

- Networked devices with limited abilities, e.g.:
 - computerized sensors (agriculture, health care, transport control, ...)
 - home controllers (house automation),
 - ...
- IoT communication – primarily wireless:
- Short-range comm. technologies:
 - WiFi,
 - BlueTooth,
 - ZigBee (IEEE 802.15.4),
 - NFC, RFID,
 - Z-Wave
 - ...

IoT Networks – Long Distance Communication

- Medium-range:
 - 5G,
 - LTE variants.
- Long-range:
 - LoRaWAN,
 - SigFox,
 - NB-IoT,
 - VSAT (satellite technology).

IoT Security

- IoT devices:
 - Limited power feeding,
 - Limited computing and storage capacity:
 - Resulting in (possible) security flaws.
- Cloud-based solutions prevail: internet communication required.
 - Confinement into a separated VLAN does not improve security much.
 - Still, certain level of separation (from private LANs) is necessary.
 - Keep an eye on authentication:
 - Weak passwords are frequent (defaults should be changed ASAP).
 - ...

Questions? Comments?

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Thank you for your attention

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